



Adaptive Kriging Particle Filter and its Application to Terrain-Aided Navigation

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Context: autonomous navigation for UAVs

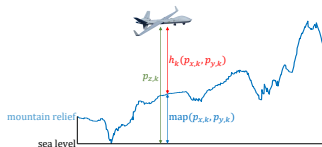
Inertial navigation for Unmanned aerial vehicles (UAVs)

- Integrates measurements from the accelerometers and gyrometers of an inertial measurement unit (IMU)
 - Autonomous and reliable on short missions
 - But imperfect measures (biases, scale factors, noises, ...)
- ⇒ Drifts over time

Data fusion

- Using data from other sensors helps reduce the drift
 - GNSS as an ideal counterpart to inertial data, but not always available
- ⇒ Physical field-aided navigation instead
- Measures from exteroceptive onboard sensors
 - Comparison between the measured data and an embedded map

Case of study: Terrain aided-navigation (TAN)

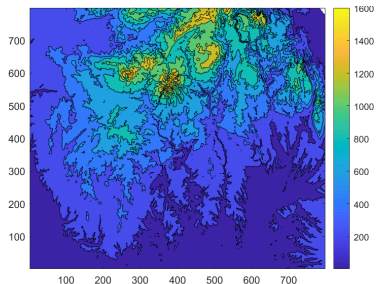


Terrain-aided navigation

- Measurement of the overflowed terrain altitude by a radar-altimeter
- Comparison with an embedded map

Challenges

- High initial state uncertainty
- Ambiguous and non-analytical environment model
- Poorly-known environment:
 - Noisy and under-sampled field
 - Better map too big to embed



Bayesian estimation

Discrete state-space representation

- $$\begin{cases} x_k = \mathcal{F}_k(x_{k-1}, \nu_k) & \text{recursive dynamical process} \\ z_k = \mathcal{H}_k(x_k) + \eta_k & \text{measurement equation} \end{cases} \quad (1)$$

Bayesian filtering

- Goal: compute $p(x_k | z_{1:k})$ the posterior probability density function (PDF)
- State estimation:

$$\hat{x}_k^{EAP} = \mathbb{E}(x_k | z_{1:k}) = \int x_k p(x_k | z_{1:k}) dx_k \quad (2)$$

- Error covariance matrix:

$$P_k^{EAP} \triangleq \mathbb{V}(\hat{x}_k^{EAP}) = \int (x_k - \hat{x}_k^{EAP})(x_k - \hat{x}_k^{EAP})^T p(x_k | z_{1:k}) dx_k \quad (3)$$

Bayesian estimation

Optimal Bayesian filter

- Recursive process in two steps:
 - Prediction step with Chapman-Kolmogorov's equation:

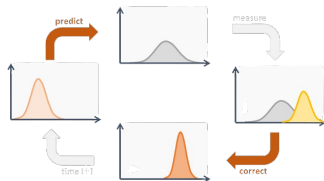
$$p(x_k | z_{1:k-1}) = \underbrace{\int p(x_k | x_{k-1}) p(x_{k-1} | z_{1:k-1}) dx_{k-1}}_{\text{no analytical solution}} \quad (4)$$

- Correction step with Bayes' formula:

$$p(x_k | z_{1:k}) = \frac{p(z_k | x_k) p(x_k | z_{1:k-1})}{\int p(z_k | x_k) p(x_k | z_{1:k-1}) dx_k} \quad (5)$$

Kalman filter

- Optimal analytical estimation under strong assumptions:
 - Linear equations and Gaussian noises
 - Poorly handles ambiguities and multi-modality
- ⇒ Stochastic methods are superior in these cases



Particle filtering

Standard particle filter

- Importance sampling method
- Asymptotically optimal with weak assumptions:
 - $\mathcal{F}_k$, $\mathcal{H}_k$ and noise distributions known
- Posterior PDF approximation using a mixture of Dirac distributions:

$$p(x_k | z_{1:k}) \approx \sum_{i=1}^{N_p} w_k^i \delta(x_k - x_k^i) \quad (6)$$

- Prediction step: $x_k^{(i)} \sim q(\cdot | x_{k-1}^{(i)}, z_k)$
- Correction step: $w_k^{(i)} \propto w_{k-1}^{(i)} \frac{p(z_k | x_k^{(i)}) p(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} \quad (7)$
- Usual proposal: $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k) = p(x_k^{(i)} | x_{k-1}^{(i)})$

Regularised particle filter (RPF)

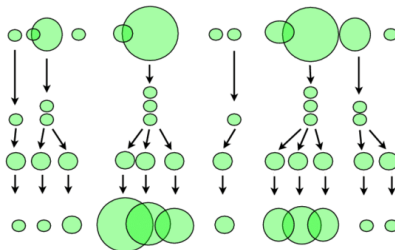
Regularised particle filter

- Particle degeneracy during the correction step → resampling if:

$$N_{\text{eff}} \triangleq \frac{1}{\sum_{i=1}^{N_p} (w_k^i)^2} < \alpha_{th} N_p \quad \text{with } 0 < \alpha_{th} < 1 \quad (8)$$

- Particle impoverishment due to duplication → regularisation
- Kernel density approximation of the PDF:

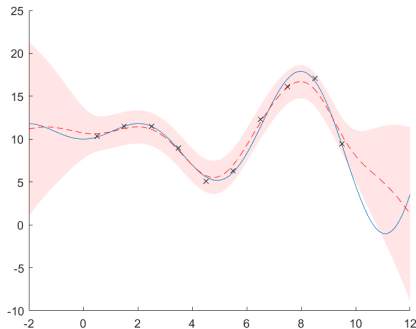
$$p(x_k | z_{1:k}) \approx \sum_{i=1}^{N_p} w_k^i \mathcal{K}_h(x_k - x_k^i) \quad (9)$$



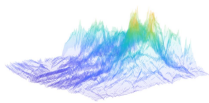
Gaussian process regression

Navigation in Poorly-Known Environment

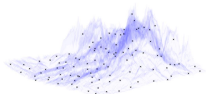
- Poorly-known environment → Bayesian correction is impossible
 - Measurement field too complex for multi-linear interpolation
- ⇒ Kriging grants an analytical expression of the measurement function and noise conditional to some training data



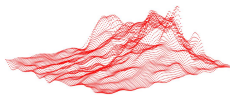
Gaussian process regression



Unknown map



Noisy and scarce training data



Reconstructed map

Gaussian process regression

- Particles and training data:

$$\begin{cases} \forall i \in \llbracket 1, n \rrbracket, & z^i = \mathcal{H}(x^i) + \eta^i & \eta^i \sim (0, R) \\ \forall j \in \llbracket 1, \tilde{n} \rrbracket, & \tilde{z}^j = \mathcal{H}(\tilde{x}^j) + \tilde{\eta}^j & \tilde{\eta}^j \sim (0, \tilde{R}) \end{cases} \quad (18)$$

- Conditional probability distribution:

$$z | \tilde{z} \sim \mathcal{N}(\mu^*(x, \tilde{x}, \tilde{z}), R^*(x, \tilde{x}))$$

$$\begin{cases} A(x, \tilde{x}) = \mathcal{K}(x, \tilde{x}) \left(\mathcal{K}(\tilde{x}, \tilde{x}) + I_{\tilde{n}} \otimes \tilde{R} \right)^{-1} \\ \mu^*(x, \tilde{x}, \tilde{z}) = \mu(x) + A(x, \tilde{x}) (\tilde{z} - \mu(\tilde{x})) \\ R^*(x, \tilde{x}) = \mathcal{K}(x, x) + I_n \otimes R - A(x, \tilde{x}) \mathcal{K}(x, \tilde{x})^T \end{cases} \quad (20)$$

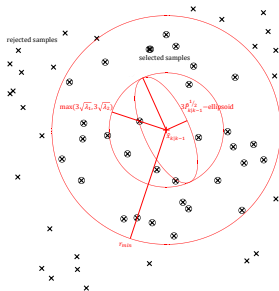
- Kriged measurement function:

$$z_k = \mu^*(x_k, \tilde{x}, \tilde{z}) + \eta_k^* \quad \eta_k^* \sim (0, R^*(x_k, \tilde{x})) \quad (21)$$

Adaptive kriging (AK)

Fitting the model

- Chosen GP: $\mu : x \mapsto \mu_0$ and $\mathcal{K} : x, x' \mapsto \sigma_f^2 \exp\left(-\frac{1}{2} \frac{(x-x')^T(x-x')}{\rho^2}\right)$
- Computation of the hyperparameters by maximum likelihood estimation



Training data selection

- Non-stationarity of the correction field
- Need to select relevant training data
- Moving window:

$$W_k \triangleq \{j \in \llbracket 1, \tilde{n} \rrbracket : \|\tilde{x}^j - \pi_g(\hat{x}_{k|k-1})\|_2 < r_k\} \quad (24)$$

- Adaptive radius:

$$\begin{cases} r_k \triangleq \min\{r \in [r_{min,k}, +\infty[: \text{Card}(W_k) \geq N_{W,min}\} \\ r_{min,k} \triangleq \alpha_W \times \max_{\lambda \in \text{sp}(\pi_g(\hat{P}_{k|k-1}))} (3\sqrt{\lambda}) \end{cases} \quad (25)$$

AK-RPF pseudo-code

Algorithm overview

$\forall i \in \llbracket 1, N_p \rrbracket$, draw x_k^i from $p(\cdot | x_{k-1}^i)$

compute r_k from (25)

extract W_k from the training set using (24)

fit the process to the extracted data by MLE

$\forall i \in \llbracket 1, N_p \rrbracket$, compute $p(z_k | x_k^i)$ according to (20)

$\forall i \in \llbracket 1, N_p \rrbracket$, compute w_k^i according to (7)

$$\hat{x}_k \leftarrow \sum_{i=1}^{N_p} w_k^i x_k^i$$

$$\hat{P}_k \leftarrow \sum_{i=1}^{N_p} w_k^i (x_k^i - \hat{x}_k)(x_k^i - \hat{x}_k)^T$$

if (8) then

resample $(x_k^{1:N_p}, w_k^{1:N_p})$

$\forall i \in \llbracket 1, N_p \rrbracket$, draw ε_k^i from the regularisation kernel

$$\forall i \in \llbracket 1, N_p \rrbracket, x_k^i \leftarrow x_k^i + \alpha_{reg} h \hat{P}_k^{\frac{1}{2}} \varepsilon_k^i$$

$$\hat{x}_k \leftarrow \sum_{i=1}^{N_p} w_k^i x_k^i$$

$$\hat{P}_k \leftarrow \sum_{i=1}^{N_p} w_k^i (x_k^i - \hat{x}_k)(x_k^i - \hat{x}_k)^T$$

end if

▷ Prediction

▷ Adaptive selection

▷ Adaptive GP

▷ Kriging

▷ Correction

▷ Estimation

▷ Resampling

▷ Regularisation

▷ Re-estimation

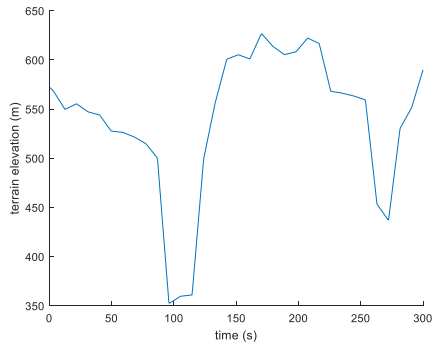
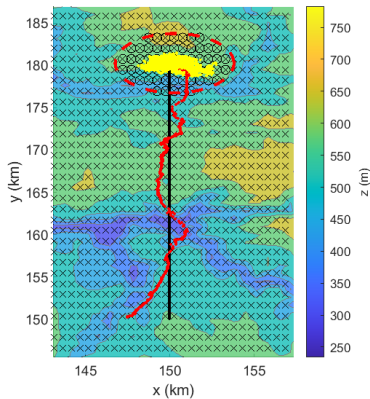
Simulation

Simulation parameters

- Uniform rectilinear motion
- 100 Monte Carlo trials per filter

Simulation	Value
Number of iterations	600
Map surface	$40 \times 100 \text{ km}^2$
Initial error covariance	$\text{diag}([2000, 2000, 100, 2, 2, 1])^2 \text{ SI}$
Evolution covariance	$\text{diag}([40, 40, 2, 0.4, 0.4, .05])^2 \text{ SI}$
Measurement covariance	15^2 m^2
Number of training data	$\in \{20^2, 30^2, \dots, 80^2\}$
Training measurement covariance	15^2 m^2
AK-RPF	Value
N_p	5000
α_{th}	0.5
α_{reg}	0.3
$N_{W,min}$	10
α_W	2

Example of an estimated trajectory



Numerical results

Comparison between AK-RPF and RPF

Resolution	RPF		AK-RPF	
	Convergence	Final RMSE	Convergence	Final RMSE
0.40km	92%	0.97km	/	/
0.70km	34%	1.31km	/	/
0.78km	21%	1.61km	/	/
0.88km	3%	3.91km	97%	0.43km
1.00km	0%	N/A	81%	1.08km
1.17km	0%	N/A	72%	1.15km
1.40km	/	/	55%	1.19km
1.75km	/	/	26%	2.35km
2.33km	/	/	1%	2.24km
3.50km	/	/	0%	N/A

RMSE as a function of time

